

Sara Sharma

AI/ML Engineer | Quantitative Analytics | Financial & Economic Data Analysis

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- Quantitative and AI/ML professional with experience in financial and economic data analysis, predictive modeling, and decision intelligence through work at **NITI Aayog, IIT Bombay, DRDO, and industry research environments**.
- Analyzed PMMY, PLFS, and Udyam datasets to study credit access, loan disbursement, employment, skills, wages, and women’s entrepreneurial ecosystems.
- Currently volunteering for DRDO, as an AI/ML researcher.

WORK EXPERIENCE

Indian Institute of Technology Bombay	Mumbai
AI/ML Intern	Sept 2026 - Present
Contribute to end-to-end machine-learning research for spatial scene understanding in architectural environments, covering dataset preparation, model development, and performance evaluation.	
NITI Aayog, Government of India	New Delhi
Data Management & Analysis Intern	Aug 2026 - Sept 2026
- Analyzed PMMY datasets covering loan sanctions, disbursements, borrower categories, regional trends, and average loan size, alongside PLFS and Udyam data, to evaluate women’s financial and entrepreneurial ecosystem. - Contributed to an AI-powered decision intelligence platform integrating 16 government datasets on credit, repayment, employment, wages, skills, education, and enterprise registration. - Applied Logistic Regression and Random Forest to assess entrepreneurial trajectories, identify potential risks, and support targeted policy interventions through data-driven decision-making.	
Fluentgrid Limited	Visakhapatnam
Python & Machine Learning Intern	Jun 2025 - Sept 2025
- Developed a multi-engine XGBoost-based forecasting system for supply-chain intelligence, supporting demand forecasting, risk analysis, and scenario analysis. - Processed operational datasets and engineered predictive features to identify patterns relevant to supply-chain performance and potential business risks. - Applied SHAP-based model interpretation to explain feature contributions and improve the transparency of analytical outputs.	
Inferrix Greentech Pvt. Ltd.	Noida
AI Research Intern	Jul 2024 - Sept 2024
- Developed a physics-informed neural network integrating HVAC, weather, building, and IoT data for predictive building-energy modeling. - Generated heat-load profiles and evaluated system behavior under varying operating and environmental conditions. - Reported a 15% reduction in building energy consumption through the proposed predictive energy-modeling approach.	

KEY PROJECTS

Machine Learning Reliability Under Distribution Shift
- Developed a framework to evaluate ML reliability under temporal and spatial distribution shift using uncertainty, calibration & OOD detection. - Compared standard, uncertainty-aware and abstention-based models to analyze failure modes and risk-performance trade-offs. - Extended evaluation to LLMs under adversarial and out-of-distribution inputs.
AI-Driven Supply Chain Intelligence Platform
- Built a 5-engine XGBoost analytics platform for forecasting, risk analysis, scenario analysis, and supply-chain decisions. - Integrated predictive outputs with SHAP-based explainability to identify influential factors and improve the interpretability of business-oriented forecasts. - Developed a web-based interface using Next.js to present analytical results and support scenario exploration.

RESEARCH & PUBLICATION EXPERIENCE

ECOAVIA: Data-Driven Aviation Demand Forecasting for Sustainable Policy Advisory ICISC 2026 (Published)	
- Developed an ensemble forecasting approach combining XGBoost, ARIMA, and LSTM models to analyze aviation-demand trends. - Integrated climate and infrastructure-related variables to support demand forecasting and sustainability policy analysis.	
Defence Research and Development Organisation Ongoing Research (Volunteer)	2026 - Present
- Conduct research under scientist supervision on AI/ML applications involving heterogeneous data modalities for defence-oriented analytical systems. - Support the development and evaluation of machine-learning approaches for integrating and interpreting complex data sources. - Apply structured experimentation and model assessment techniques to investigate performance across varied operating conditions.	

SKILLS

Programming & Data Analysis: Python, SQL, JavaScript, Pandas, NumPy, PostgreSQL, Exploratory Data Analysis, Data Cleaning, Data Validation, Data Visualization
Quantitative & Analytical Methods: Predictive Analytics, Statistical Modeling, Feature Engineering, Time-Series Modeling, Forecasting, Scenario Analysis, Risk Scoring, Model Evaluation, Model Interpretation, Data Integration, Data Harmonization
Machine Learning: Logistic Regression, Random Forest, XGBoost, Neural Networks, Deep Learning, Physics-Informed Neural Networks, Uncertainty Estimation, Out-of-Distribution Detection, Explainable AI, SHAP
Tools & Platforms: PyTorch, TensorFlow, Scikit-learn, MLflow, FastAPI, Docker, Git, CI/CD, Next.js
Domain Exposure: Financial and Economic Data Analysis, Credit and Loan-Disbursement Data, Labour-Market Indicators, Enterprise-Registration Data, Policy Research, Decision Intelligence, Evidence-Based Policy Analysis